

Harsh Gupta

+91 8299244461 | harsh16official@gmail.com | Portfolio | LinkedIn | GitHub | X

Experience

Google Summer of Code 2026

May 2026 – Present

Joplin

Remote

- Engineered an on-device, privacy-preserving note categorization plugin using Transformers.js and ONNX runtime, executing 100% of embedding computations locally to eliminate cloud latency and data risks.
- Designed a chunk-embed-average pipeline with title weighting, benchmarking embedding models to select BGE-small-en-v1.5 for optimal clustering accuracy and low memory footprint.
- Implemented semantic clustering with UMAP dimensionality reduction and HDBSCAN/K-Means algorithms, achieving a top clustering score of **0.94** across **7** auto-detected topic clusters.
- Built a tag-suggestion pipeline combining TF-IDF keyword extraction with centroid cosine-similarity re-ranking to generate descriptive multi-word cluster tags.

Full Stack Engineer Intern

Sep 2025 – Nov 2025

GRS Worker

Remote

- Architected a full-stack web application using Next.js 15, Prisma, and NextAuth, securing multi-role user sessions and modeling complex relational database schemas.
- Integrated Twilio SMS notifications, instant QR code generation, and Zod schema validation, reducing field data entry errors by **40%**.
- Built a responsive UI with Radix UI and Framer Motion, optimizing API endpoint throughput via Turbopack and Server Actions to drive **20% post-launch user growth**.

Projects

Botforweb – Documentation Chatbot SaaS | FastAPI, PostgreSQL, Qdrant, Celery, React

Live | GitHub

- Built a multi-tenant documentation-chatbot SaaS enforcing tenant isolation via PostgreSQL Row-Level Security and per-tenant Qdrant vector collections.
- Designed an async RAG ingestion pipeline with Celery/Redis task queues, hybrid dense/sparse retrieval, Cohere reranking, and DeepSeek LLM generation to deploy fact-checked site chatbots in **under 5 minutes**.

Nimicode – Terminal AI Coding Agent | Bun, TypeScript, React (Ink), DeepSeek API

Live | GitHub

- Developed a terminal-based AI coding agent CLI with an autonomous think-execute tool loop and a React/Ink interactive TUI.
- Implemented file read/write and shell-execution tool handlers with a provider-agnostic LLM layer, cutting API token expenses by **90%**.

Chess – Real-Time Multiplayer Chess | React, TypeScript, WebSockets, Prisma, PostgreSQL

Live | GitHub

- Built a real-time multiplayer chess platform with WebSocket-based move synchronization, timed matches, and cheat-proof server-side move validation.
- Integrated an AI grandmaster coach for on-demand position evaluation and added game history replay with move-by-move navigation.

Achievements

- Merged **60+ pull requests** across major open-source organizations, including Joplin and AsyncAPI.

Technical Skills

Languages: JavaScript, TypeScript, Python

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS, Framer Motion, Radix UI

Backend & Web: Node.js, Express.js, FastAPI, WebSockets, WebRTC, Bun

Databases: PostgreSQL, MongoDB, Prisma, Qdrant, Redis

AI / ML: Transformers.js, Embeddings, Vector Search, RAG Pipelines, NLP, UMAP, PyTorch, LangChain

Tools: Git, GitHub, Docker, Turborepo, Celery

Education

Harcourt Butler Technical University

B.Tech

Aug 2023 – Jun 2027

Kanpur, UP